Joonsuk Bae

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Main analyzer of the first ALICE Run 3 inclusive charged-particle jet cross section — preliminary released; paper in preparation; presented the jet working group's combined results at EPS-HEP 2025. Detector-side work from pp tracking systematics to Pb/scintillating-fiber calorimetry R&D for the ePIC Barrel Imaging Calorimeter.

Education

Ph.D. Candidate, Physics — Sungkyunkwan University 2022.03 – expected 2027.02

Advisor: Beomkyu Kim. Thesis: *R-dependent charged-particle jet production in pp collisions at $\sqrt{s} = 13.6$ TeV with ALICE.* Defense scheduled 2026.12.02; degree conferral 2027.02.

Research Assistant — Institute of Basic Science (Physics), Sungkyunkwan University 2021.03 – 2021.10

B.S., Physics — Sungkyunkwan University 2013.03 – 2021.02

Appointments and Collaboration Membership

ALICE Collaboration, CERN, Geneva	2022.10 – present
ePIC Collaboration, Brookhaven National Laboratory	2024.02 – present
LAMPS Collaboration, RAON, Korea	2022.03 – 2024.04

Research Experience

Main analyzer, Run 3 charged-particle jets — ALICE jet working group 2023.04 – present

- **First ALICE Run 3 inclusive charged-particle jet cross section**
  Responsible for the full chain: O^{2}Physics task development, QA on continuous-readout data, Bayesian/SVD unfolding, and the systematic-uncertainty program. ALICE Preliminary; paper in preparation. Varied $R = 0.2$–0.6 to separate collinear from large-angle radiation.
- **Tracking-efficiency systematics for Hard Probes 2026**
  Derived the pp charged-particle tracking-efficiency uncertainty at $\sqrt{s} = 13.6$ and 5.36 TeV (p_{T}-binned, 2022–2024 periods), combining the track-selection component with the ITS–TPC matching component.
- **Track momentum resolution**
  Studied data–MC resolution mismatch by covariance propagation, MC residuals, and V0 mass resolution, and derived the p_{T}-dependent smearing profile used in Run 3 charged-jet corrections and systematics.
- **Cross-section normalization**
  Built the Run 3 charged-jet normalization now used across the group, from the ALICE TVX visible cross section under continuous readout, quantifying the bunch-crossing and track-matching inefficiencies of time-frame-based reconstruction and deriving the corresponding signal-loss corrections. Contributed the cross-section normalization of the GNN-tagged b-jet cross-section analysis in pp collisions at $\sqrt{s} = 13.6$ TeV.
- **Event-generator production for model comparison**
  Produced and managed the comparison sets behind ALICE jet measurements, spanning LO tune variations (PYTHIA 8: Monash, colour reconnection, rope hadronization, string shoving), NLO matching (POWHEG BOX), and an independent parton-shower and hadronization model (HERWIG 7).

Independent study, machine learning at high occupancy — public FM4NPP model 2026.08 – present

- Designed the question and the evaluation; reproduced the published noise-hit tagging result of the FM4NPP foundation model (arXiv:2508.14087) on the TPCpp-10M benchmark within one percentage point, then raised the hit density by overlaying p+p events. At 98.1% real-hit efficiency, noise removed by the p+p-trained tagger falls from 85% to 11% at a hit density comparable to 20–30% central Au+Au (100 overlaid p+p events); a per-hit head retrained at that density recovers only part of it.

Detector physicist, Barrel Imaging Calorimeter (BIC) — Korea-BIC group, ePIC 2024.02 – present

- **Pb/scintillating-fiber prototype test beams**
  Lead analyzer for the CERN PS T10 (2025.07) campaign: energy response, resolution, and linearity. On-site DAQ shifter with equalization and prompt analysis at KEK AF–AR (2025.03), and at the first PS T10 campaign (2024.08), co-author of the resulting paper.
- **Module QA and assembly**
  Ran the PMT-module QA for the prototype modules before the test beams (cosmic-ray and source characterization, timing-coincidence checks, optical-coupling stability, per-module acceptance criteria); Pb/scintillating-fiber bundle alignment, polishing, and PMT/SiPM coupling.
- Energy- and η-dependent sampling-fraction studies in JANA2 / EIC-recon for the Korea-BIC simulation effort; Geant4 prototype modeling.

Detector R&D, Forward Tracker — LAMPS, RAON 2022.03 – 2024.04

- Early-stage design and Geant4 simulation of an MPPC–scintillator-plane forward tracking detector; sensor surveys, specifications, and tracking-concept feasibility studies.

Selected Publications

INSPIRE: inspirehep.net/authors/2117232 · ORCID: 0009-0008-4806-8019. ALICE Collaboration co-author since 2022.10; ECCE/ePIC and LAMPS collaboration papers are included in the INSPIRE record.

- ALICE Collaboration, *R-dependent charged-particle jet production in pp collisions at $\sqrt{s} = 13.6$ TeV with ALICE* (submission expected 2026). Thesis paper.
  [main analyzer]
- H. Lee, C. Lee, J. Ryu, G. An, **J. Bae** *et al.*, "Beam test of a Pb/SciFi prototype for the Barrel Imaging Calorimeter at the Electron-Ion Collider" — CERN PS T10 (2024.08) campaign; accepted for publication in *J. Korean Phys. Soc.*; `arXiv:2604.22647`.
  [co-author; module-construction support and PMT/module performance characterization]
- Y. Hong, J. Bok, G. An, **J. Bae** *et al.*, "Test-beam performance of the AstroPix silicon sensor for imaging calorimetry," accepted for publication in *Nucl. Instrum. Meth. A*; `arXiv:2605.07681`.
  [co-author; beam-test DQM shifts and on-site prompt analysis]
- S. Y. Jang, G. An, **J. Bae** *et al.*, "Performance of dual-readout calorimeters for various absorbers," *Nucl. Instrum. Meth. A* (2026), DOI `10.1016/j.nima.2026.171598`.
  [co-author; beam-test participation and DAQ shifts]
- G. Cho, S. Kim, **J. Bae** *et al.*, "Beam tests of the copper-based dual-readout calorimeter to measure electromagnetic performance for future e^+e^- colliders," *J. Subatom. Part. Cosmol.* **3** (2025) 100021.
  [co-author; module construction and beam-test data taking]
- Y. J. Kim, D. S. Ahn, J. K. Ahn *et al.* (LAMPS), "Large Acceptance Multi-Purpose Spectrometer (LAMPS) at the RAON," *J. Korean Phys. Soc.* **87** (2025) 670–682.
  [co-author; Time-of-Flight detector maintenance and on-site checks]
- Y. Bae *et al.* (LAMPS), "Performance test of prototype LAMPS Start Counter using proton beams at 100 MeV," *Nucl. Instrum. Meth. A* (2025), DOI `10.1016/j.nima.2025.170827`.
  [co-author; Start-Counter prototype testing]
- Y. Cheon *et al.* (LAMPS), "Characteristics of the prototype LAMPS TPC for low-energy nuclear collisions at RAON," *Nucl. Instrum. Meth. A* **1066** (2024) 169610.
  [co-author; TPC prototype characterization]
- H. Kim *et al.* (LAMPS), "Performance of the prototype beam drift chamber for LAMPS at RAON with proton and ^{12}C beams," *JINST* **19** (2024) P12008; `arXiv:2412.08662`.
  [co-author; forward-tracking R&D and prototype performance studies]

Invited and Contributed Talks

Invited Talks (external workshops, seminars)

- **France–Korea Particle Physics Network (FKPPN) Workshop**, Inha University — 2025.12
  Cross-section normalization in Run 3; charged-jet studies in ALICE Run 3
- **1st Heavy-Ion Meeting (HIM 2025)**, APCTP — 2025.05
  From precision to structure: jet measurements in ALICE Run 3 and ongoing substructure studies
- **FKPPN Workshop**, Inha University — 2023.11
  Charged-jet studies in ALICE Run 3

Collaboration-Meeting and National-Workshop Talks (internal)

- **ALICE Physics Forum**, Hard Probes 2026 approvals — 2026.06
  R-dependence of inclusive charged-particle jet spectrum in pp (with M. Kim)
- **Korea-BIC 2025 Summer Workshop** — 2025.09
  Data analysis of the recent beam test at CERN
- **KoALICE National Workshop 2024** — 2025.01
  Jet analysis status: first look at the charged-particle jet spectrum in Run 3 pp collisions
- **ALICE Physics Week 2024** — 2024.12
  Charged-particle jet production in Run 3
- **ALICE Physics Forum**, Hard Probes 2024 approvals — 2024.09
  Inclusive pp jet spectrum in Run 3
- **KoALICE National Workshop 2023** — 2024.01
  Charged-jet O^2 MC and data QA
- **KoALICE National Workshop 2022** — 2023.01
  Dijet studies at the LHC

Contributed Talks and Posters — International Conferences

- **Hard Probes 2026**, Nashville *[talk]* — 2026.06
  Jet-like dihadron correlations in pO, OO, and Ne–Ne collisions
- **Hard Probes 2026**, Nashville *[poster]* — 2026.06
  Characterizing inclusive charged-particle jet production and fragmentation in pp collisions
- **EPS-HEP 2025**, Marseille *[jet working group's combined results]* — 2025.07
  Inclusive and semi-inclusive jet cross-section measurements in pp collisions
- **INPC 2025**, Daejeon *[talk]* — 2025.05
  Charged-particle jet and jet quenching measurement in pp collisions
- **Quark Matter 2025**, Frankfurt *[poster]* — 2025.04
  Inclusive charged-particle jet spectrum in pp collisions in Run 3
- **Hard Probes 2024**, Nagasaki *[poster]* — 2024.09
  First measurements of charged-particle jet production in pp collisions
- **Quark Matter 2023**, Houston *[poster]* — 2023.09
  Charged-particle multiplicity distribution in pp collisions
- **ATHIC 2023** (9th Asian Triangle Heavy-Ion Conf.), Incheon *[poster]* — 2023.04
  Dijet studies at the LHC

Contributed Talks and Posters — Korean Physical Society

- **KPS Fall 2026** *[poster]* — 2026.10
  Occupancy stress test of the nuclear-physics foundation model FM4NPP on the TPCpp-10M open benchmark
- **KPS Spring 2026** *[talk]* — 2026.04
  R-dependent charged-particle jet production in pp collisions

- **KPS Fall 2025** *[talk]* — 2025.10
  Inclusive and semi-inclusive jet cross-section measurements in pp collisions
- **KPS Spring 2025** *[talk]* — 2025.04
  Inclusive charged-particle jet spectrum in pp collisions in Run 3
- **KPS Spring 2025** *[poster]* — 2025.04
  Prototype calorimeter QC for the Barrel Imaging Calorimeter at the EIC
- **KPS Fall 2024** *[talk]* — 2024.10
  First jet measurement in ALICE Run 3: charged-particle jet production
- **KPS Spring 2024** *[talk]* — 2024.04
  Charged-particle jet production in pp collisions
- **KPS Fall 2023** *[talk]* — 2023.10
  First look at charged-particle jet production in pp collisions
- **KPS Spring 2023** *[poster]* — 2023.04
  Dijet studies at the LHC
- **KPS Fall 2022** *[poster]* — 2022.10
  Dijet studies at the LHC

Service and Professional Activities

- **ALICE Run 3 O+O charged-jet measurement**
  One of the principal authors; derived the cross-section normalization and its QA; provided the POWHEG-based pp reference for the R_{OO} preliminary.
- **ePIC Korea-BIC beam-test analysis framework**
  Developed and shared the analysis framework across the Korean BIC institutes (SKKU, PNU, KNU).
- **ALICE ITS2 alignment monitoring** (Run 3)
  Service task: period-by-period tracking-performance diagnostics from compressed-time-frame data.
- **KoALICE national activities**
  Regular participation since 2022, including invited collaboration-meeting talks.
- **France–Korea ALICE activities**
  FKPPN workshop every year since 2023; weekly ALICE-Asian-France (AAF) group discussion.

Awards and Fellowships

- **BK21 FOUR Graduate Scholarship**
  Awarded on research performance by the Sungkyunkwan University Physics BK21 sponsored unit (Korean Ministry of Education), 2022 – present.

Teaching and Mentoring

Student mentoring (non-supervisory) — Sungkyunkwan University

- *Wooseok Ham* (M.Sc., main analyzer of the O+O charged-jet measurement) — 2024 – present
  Mentored on Run 3 jet reconstruction methodology, correction procedures, and systematic-uncertainty estimation; jointly authored the corresponding ALICE Analysis Note.
- *Harim Seong* (undergraduate) — 2024.03 – 2024.11
  ePIC BIC detector-performance studies using JANA2-based reconstruction; energy response and sampling-fraction extraction.

KoALICE O^2 Tutorial — student co-organizer and instructor — Sungkyunkwan University 2024.01

- Co-organized (with H. Lee) the two-day Run 3 analysis-framework tutorial for the Korean ALICE community (22 participants); instructor on *Running O^2 Physics with Hyperloop.*

Teaching Assistant — Department of Physics, Sungkyunkwan University 2022 – 2024

Nuclear Reactions (graduate), *Quantum Mechanics II*, *General Physics I and Laboratory I.*

Technical Skills

Languages. English (professional working proficiency), Korean (native).
Programming. C++, Python, ROOT, LaTeX, Bash, Git.
HEP frameworks. ALICE O^{2}Physics and Hyperloop (grid analysis trains), Geant4, JANA2 / EIC-recon, FastJet.
Software contributions. ALICE O^{2}Physics — 22 merged pull requests (2023–2026): C++ analysis tasks on the framework's columnar data model, and a fix to the common track-smearing tool (TrackTuner).
Event generators. PYTHIA 8, POWHEG BOX, HERWIG 7.
Analysis methods. jet reconstruction and unfolding (Bayesian, SVD), efficiency and systematic-uncertainty evaluation, data/MC validation, test-beam calibration and energy-response fitting.
Detector hardware. PMT and SiPM readout characterization, cosmic-ray and source calibration, optical-coupling QA, calorimeter module assembly.